This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.

Designation: E1582 – 25

Standard Test Method for Temperature Calibration of Thermogravimetric Analyzers[1]

This standard is issued under the fixed designation E1582; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

1. Scope*

1.1 This test method describes the temperature calibration of thermogravimetric analyzers over the temperature range from 25 °C to 1500 °C and is applicable to commercial and custom-built apparatus. This calibration may be accomplished by the use of either melting point standards or magnetic transition standards.

1.2 The weight change curve in thermogravimetry results from a number of influences, some of which are characteristic of the specimen holder assembly and atmosphere rather than the specimen. The variations from instrument to instrument occur in the point of measurement of the temperature, the nature of the material, its size and packing, the geometry and composition of the specimen container, the geometry and design of the furnace, and the accuracy and sensitivity of the temperature sensor and displaying scales. These all contribute to differences in measured temperatures, which may exceed 20 °C. In addition, some sample holder assemblies will show variations of measured temperature with sample size or heating/cooling rate, or both. Since it is neither practical nor advisable to standardize sample holders or thermobalance geometries, instruments may be calibrated by measurement of the deviation of a melting or magnetic (Curie point) transition temperature from the standard reference temperature. This deviation can be applied as a correction term to subsequent measurements.

1.3 This test method assumes that the indicated temperature of the instrument is linear over the range defined by a two-point calibration and that this linearity has been verified. These two calibration temperatures should be as close to the experimental measurements to be made as possible.

1.4 This test method describes two procedures for temperature calibration of thermogravimetric analyzers using any type balance. Procedure A uses melting point standards for calibration. Procedure B uses magnetic transition standards for calibration.

1.5 The data generated by these procedures can be used to correct the temperature scale of the instrument by either a positive or negative amount using either a one- or two-point temperature calibration procedure.

1.6 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.7 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.*

1.8 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

2. Referenced Documents

2.1 *ASTM Standards:*[2]

E473 Terminology Relating to Thermal Analysis and Rheology

E691 Practice for Conducting an Interlaboratory Study to Determine the Precision of a Test Method

E967 Test Method for Temperature Calibration of Differential Scanning Calorimeters and Differential Thermal Analyzers

E1142 Terminology Relating to Thermophysical Properties

E2040 Test Method for Mass Scale Calibration of Thermogravimetric Analyzers

E3142 Test Method for Thermal Lag of Thermal Analysis Apparatus

3. Terminology

3.1 *Definitions:*

3.1.1 Technical terms used in this document are defined in Terminologies E473 and E1142, including *Celsius, Curie*

[1] This test method is under the jurisdiction of ASTM Committee E37 on Thermal Measurements and is the direct responsibility of Subcommittee E37.01 on Calorimetry and Mass Loss.

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[2] For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer service at www.astm.org/contact. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

*A Summary of Changes section appears at the end of this standard

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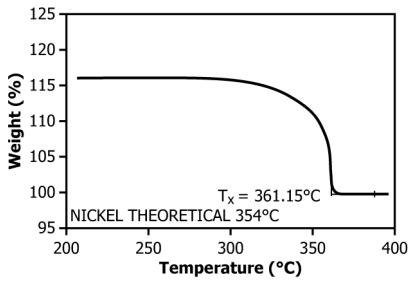

FIG. 1 Magnetic Reference Temperature

temperature, derivative, Kelvin, magnetic transformation, onset point, thermogravimetry, thermogravimetric analyzer, and *thermomagnitometry*.

3.2 *Definitions of Terms Specific to This Standard:*

3.2.1 *magnetic reference temperature, n*—the observed temperature at which a change in the magnetic properties of a material in a magnetic field produces an apparent mass change. This temperature is read from the dynamic thermogravimetry curve as the point of intersection of the extrapolated higher temperature portion of the base line with a tangent drawn to the point of greatest slope (inflection point) of apparent mass-change curve. This temperature most closely represents the Curie point, that point on the mass change curve where the magnetic effect of the standard material has disappeared completely (see Fig. 1).

3.2.1.1 *Discussion*—The position of the magnet and the design of the instrument will affect the direction of the mass change.

4. Summary of Practice

4.1 This test method provides a set of different procedures since thermogravimetric apparatus are often of significantly differing designs.

4.2 *Calibration of Analyzers Using Magnetic Transition Standards:*

4.2.1 In this procedure, the apparent mass change of one or more of the magnetic transition standards is obtained under the normal operating conditions of the instrument. The extrapolated endpoint temperature (see Fig. 1) is determined and compared with the established transition temperature for the material. The difference provides an adjustment or calibration that may be applied to the temperature scale of the instrument.

4.2.2 The apparent mass change of the magnetic transition materials is caused by the magnetic (ferromagnetic) to nonmagnetic (paramagnetic) transition in the presence of a magnetic field. This process is known as thermomagnetometry.

4.3 *Calibration of Analyzers That Have Simultaneous Thermogravimetry-Differential Scanning Calorimeter or Thermogravimetry-Differential Thermal Analysis Capability*—These instruments may be calibrated using melting temperature standards following Practice E967.

5. Significance and Use

5.1 Thermogravimetric analyzers are used to characterize a broad range of materials. In most cases, one of the desired values to be assigned in thermogravimetric measurements is the temperature at which significant changes in specimen mass occur. Therefore, the temperature axis (abscissa) of all apparent-mass-change curves must be calibrated accurately, either by direct reading of a temperature sensor, or by adjusting the programmed temperature to match the actual temperature over the temperature range of interest. In the latter case, this is accomplished by the use of either melting point or magnetic transition standards.

5.2 This test method permits interlaboratory comparison and intralaboratory correlation of instrumental temperature scale data.

6. Interferences

6.1 The reference metals are sensitive to impurities and may oxidize at elevated temperatures. All runs shall be conducted in an oxygen-free inert purge gas of the same type to be used in the experimental procedures.

6.2 Care must be taken to stay below temperatures at which the magnetic transition standard will react with the specimen or its holder.

6.3 The atmosphere, purge gas type, purge gas flow rate, and heating rate will affect the calibration. These rates and conditions must be the same for both calibration and analysis. In addition, high heating rates should be avoided, if possible. Due to the differing heat exchange (emissivity and heat capacity) during the calibration and analysis, higher heating rates increase the error in the temperature measurement.

7. Apparatus

7.1 *Thermogravimetric Analyzer*—A system of related instruments that are capable of continuously measuring the mass of a specimen in a controlled atmosphere and in a controlled temperature environment ranging from ambient to at least 25 °C above the temperature range of interest over a selected time period. This instrument shall consist of the following:

7.1.1 *Thermobalance,* composed of:

7.1.1.1 *Furnace,* to provide uniform controlled heating of a specimen from 25 °C to a constant temperature or at a constant rate within the applicable temperature range of this test method.

7.1.1.2 *Temperature Sensor,* to provide an indication of the specimen/furnace temperature to ±0.1 °C.

7.1.1.3 A continuously recording *Balance,* to measure the specimen apparent mass with a minimum capacity of 50 mg and a sensitivity of ±5 µg.

7.1.1.4 A means of maintaining the specimen/container under *Atmospheric Control,* of nitrogen or other inert gas of 99.9+ % purity at a purge rate of 10 mL/min to 100 mL/min constant to within ±5 mL/min.

7.1.2 A *Temperature Controller,* capable of executing a specific temperature program by operating the furnace between selected temperature limits at a specified heating rate between 0.5 °C/min to 20 °C/min constant to within ±0.1 °C/min or to an isothermal temperature that is maintained constant to ±0.5 °C for a minimum of 10 min.

7.1.3 A *Data Collection Device,* to provide a means of acquiring, storing, and displaying measured or calculated

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TABLE 1 Recommended Melting Temperature Standards

Calibration Material	Melting Temperature, °C (K)
Indium[A]	156.5985 (429.7485)
Tin[A]	231.928 (505.078)
Zinc[A]	419.527 (692.677)
Aluminum[A]	660.323 (933.473)
Silver[A]	961.78 (1234.93)
Gold[A]	1064.18 (1337.33)
Copper[A]	1084.62 (1357.77)
Nickel[B]	1455 (1728)
Palladium[B]	1554.8 (1828.0)
Platinum[B]	1768.2 (2041.3)

[A] Primary fixed points, ITS-90 (**1**).
[B] Secondary reference points, ITS-90 (**2**).

TABLE 2 Curie Temperature Standards[A,B,C]

Metal	Curie Point (Magnetic) Transition, °C
Alumel[A,C]	153.9
Nickel[B]	358.2
Nickel (83 %)–Cobalt (17 %)	554.4
Nickel (63 %)–Cobalt (37 %)	746.4
Iron[C]	771.4
Nickel (37 %)–Cobalt (63 %)	930.8
Cobalt	1116

[A] Available from TA Instruments, 109 Lukens Drive, New Castle, DE., the only known source of certified and traceable Curie Temperature Reference Materials. If you are aware of alternative suppliers, please provide this information to ASTM International Headquarters. Your comments will receive careful consideration at a meeting of the responsible technical committee,[1] which you may attend.
[B] From Ref. (**4**).
[C] From Ref. (**5**).

signals, or both. The minimum output signals required for thermogravimetry are weight, temperature, and time.

7.1.4 *Containers (pans, crucibles, and the like),* that are inert to the specimen and will remain dimensionally stable within the temperature limits of this test method.

7.2 For the Magnetic Transition Method, a means of applying a *magnetic field* sufficient to produce a 2 % weight change in the test specimen.

8. Calibration and Standardization

8.1 *Calibration of Apparatus*—If necessary, calibrate the mass base and temperature sensors of the instrument at room temperature using the procedure described in the instrument manual or Test Method E2040.

8.2 *Calibration Materials:*

8.2.1 *Melting Point Standards*—For the temperature range covered by many applications, the melting transition of the 99.9+ % pure materials listed in Table 1 may be used for calibration.

NOTE 1—The values in Table 1 were determined using special 99.9999 % pure materials and highly accurate steady-state conditions that are not attainable with this test method. The actual precision of this test method is given in Section 13.

8.2.2 *Magnetic Transition Standards.*

NOTE 2—Materials with known magnetic transitions determined with high precision are required (**3**).[3] Sources of materials of known or certified Curie transition temperatures can usually be sourced through thermogravimetric instrument companies. The values for Curie transition temperatures differ from lot to lot of the material. Curie point temperatures given in Table 2 were obtained from Ref. (**4**).

9. Procedure A—Melting Point Standard Calibration

9.1 *Positioning of the Temperature Sensor (Thermocouple):*

9.1.1 For some instruments, the temperature sensor is an integral part of the sample holder and cannot be adjusted. For those instruments that have adjustable temperature sensors, it shall be adjusted so that it is just beneath (but not touching) the thermogravimetric analyzer specimen container (see Figs. 4 and 5).

9.2 *Standard Preparation Procedure:*

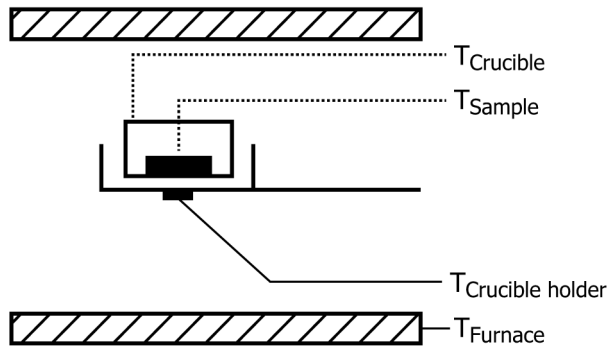

FIG. 2 Sample Positioning for Procedure A

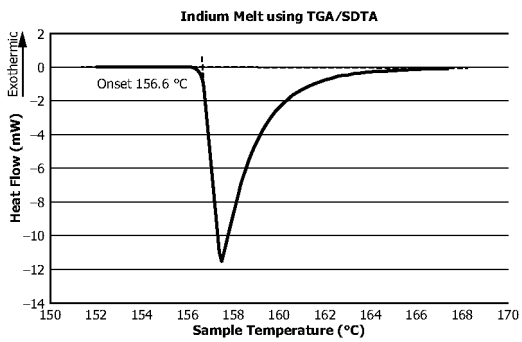

FIG. 3 Heat Flow Curve for Procedure A

9.2.1 Press the metal standard flat and place it in the middle of the crucible. Place the sample/crucible on the center of the sample holder (sensor), see Fig. 2.

9.2.2 Close the balance assembly. Purge the balance and furnace tube with the desired atmosphere and at the selected flow rate. Select the heating rate that will be used in subsequent analyses. See 6.3.

9.2.3 Adjust the balance so that it now gives a zero mass signal.

9.2.4 Close the system.

9.2.5 Rapidly heat the system to 50 °C below the theoretical melting temperature, and allow the system to equilibrate for 5 minutes. Then heat at the selected programmed rate up through the melting temperature of the standard. When the standard melts, the heat flow signal will show and endothermic peak as

[3] The boldface numbers in parentheses refer to a list of references at the end of this standard.

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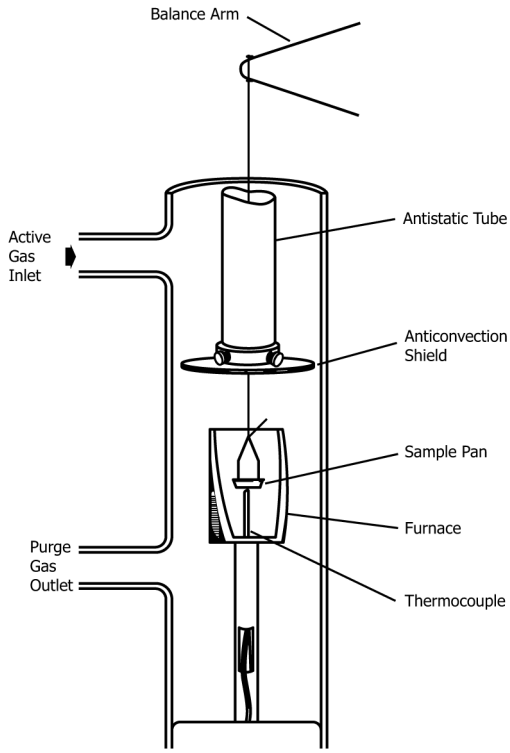

FIG. 4 Positioning of the Temperature Sensor (Thermocouple)

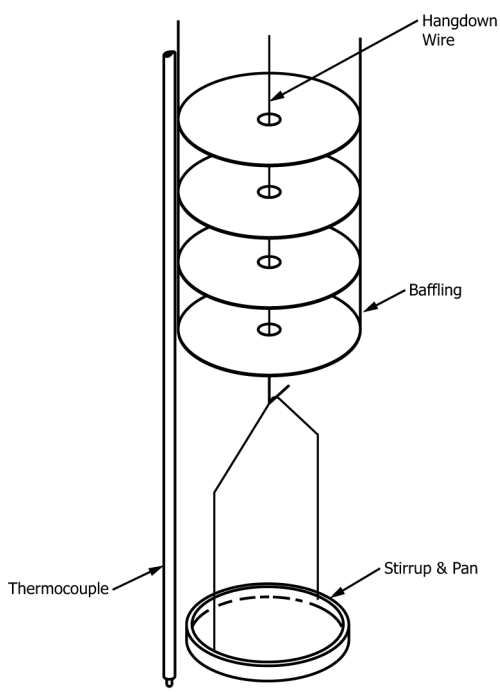

FIG. 5 Positioning of the Temperature Sensor (Thermocouple)

is typically seen with a traditional DSC or DTA. A typical curve is shown in Fig. 3.

9.2.6 Measure the onset temperature of the endothermic heat flow peak.

10. Procedure B—Magnetic Transition Standard Calibration

10.1 For those systems using internal microfurnaces with an adjustable temperature sensor, it shall be adjusted so that it is just beneath (but not touching) the thermogravimetric analyzer specimen container (see Figs. 4 and 5).

10.2 Close the system, adjust the atmospheric flow rate to the selected rate, and zero the balance and chart recorder if used.

10.3 Open the system and place a specimen of the magnetic transition material in the specimen container in the same position as that in which one would place a specimen. Close the system.

10.4 Place a permanent magnet or electromagnet outside the furnace in a position as close to the furnace and specimen container as is practical in order to affect an apparent mass change at the reference material transition temperature. Remove (or turn off) the magnet to ascertain whether a conveniently measurable apparent mass change occurs.

NOTE 3—The choice of magnets or magnetic fields should be so that an apparent mass change of up to 2 % is observed by turning the field on and off.

10.4.1 Adjust the specimen size or mass loss sensitivity, if necessary, and replace (or turn on) the magnet.

10.5 Rapidly heat the magnetic transition standard to a temperature at least 50 °C below the transition temperature of the standard and hold that temperature for 5 min.

10.6 Heat the test specimen at the desired heating rate through the transition temperature region until an apparent mass change has been observed (T_x, Fig. 1).

NOTE 4—To allow full ramp equilibration prior to the transition, the scanning ramp should extend at least 5 min prior to the observed transition (hence, the 50 °C mentioned would be appropriate for scanning rates below 10 °C/min).

NOTE 5—More than one reference material can be used in one run. Additionally, if the magnetic reference temperature of the reference material is sufficiently different from the mass change region of a test substance that the apparent and real mass changes do not interact, the reference material may be used during measurements on the test substance. Chemical interaction of any type with the sample may alter the temperature of the magnetic transition.

NOTE 6—The same specimen container and heating rates should be used in both the calibration procedures and the analytical determinations. See 6.3.

10.7 Determine the magnetic transition temperature by observing that point on the mass change curve that most closely agrees with that of T_x of Fig. 1. Computer or hand-made determination of this point can be made.

11. Calculation

11.1 For the purposes of these procedures, it is assumed that the relationship between observed temperature (TO) and actual specimen temperature (T) is a linear one governed by the following equation:

$$T = (TO \times S) + I \quad (1)$$

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where:

S and I = slope and intercept, respectively.

(See 11.2 for the means to calculate S and I, used in (Eq 1).)

NOTE 7—For some instruments, the assumption of a linear relation between observed and actual specimen temperature may not hold. Under such conditions, calibration temperatures sufficiently close together shall be used so that the instrument calibration is achieved with a series of linear relations.

11.2 *Two-Point Calibration:*

11.2.1 Using the standard temperature values taken from Table 1 and Table 2 and the corresponding observed temperatures taken from the experimental sections above, calculate the slope and intercept using the following equations:

$$S = (TS_1 - TS_2)/(TO_1 - TO_2) \quad (2)$$

$$I = [(TO_1 \times TS_2) - (TS_1 \times TO_2)]/(TO_1 - TO_2) \quad (3)$$

where:

S = slope (nominal value = 1.00),
I = intercept,
TS_1 = reference transition temperature for Standard 1 from Table,
TS_2 = reference transition temperature for Standard 2 from Table,
TO_1 = observed transition temperature for Standard 1 determined in either Section 9 or 10, and
TO_2 = observed transition temperature for Standard 2 determined in either Section 9 or 10.

NOTE 8—I has the same units (that is, °C or K) as TS_1, TS_2, TO_1, and TO_2, which are consistent with each other. The value for I will be different, depending on the units used. S is a dimensionless number whose value is independent of the units of I and T.

11.2.2 When performing these calculations, retain all available decimal places in the measured value and in intermediate values in the calculation such as the values for S and I. The final calculated or corrected temperature should be rounded to the decimal place equivalent to two significant places in the standard deviation. If, however, this temperature is to be used in a subsequent calculation, all available decimal places should be retained.

11.3 *One-Point Calibration*—If the slope value determined in 11.2.1 is between 0.999 and 1.001, only the intercept need be determined through a one-point calibration procedure.

$$I = TS_1 - TO_1 \quad (4)$$

11.4 Using the determined values for S and I (Eq 1) may be used to calculate the actual specimen transition temperature (T) from an observed transition temperature (TO).

12. Report

12.1 Report the following information:

12.1.1 Complete identification and description of the standard reference materials used for calibration, including source and purity.

12.1.2 Model number and description of the instrument used for calibration, including the location of the thermocouple.

12.1.3 Details of the procedure used for calibration, including description of the type of specimen boat or pan used and any departures from the described procedure.

12.1.4 Identification of the specimen atmosphere by gas composition, flow rate, and purity.

12.1.5 Heating rate used.

12.1.6 Statement of the dimensions, geometry, and materials of the specimen holder; and the method of loading the specimen.

12.1.7 A copy of all original records that are presented.

12.1.8 The specific dated version of this test method used.

13. Precision and Bias

13.1 The precision of magnetic transition method was determined by an interlaboratory test in 1989 with the statistical analysis conducted in accordance with Practice E691. A single material was used for the interlaboratory test. Four laboratories participated in the magnetic transition procedure.[4]

13.2 *Precision (Magnetic Transition Method):*

13.2.1 Within laboratory variability may be described using the repeatability value (r) obtained by multiplying the repeatability standard deviation by 2.8. The repeatability value estimates the 95 % confidence limit. That is, two results obtained in the same laboratory, using the same apparatus, on the same specimen, closely spaced in time have a 95 % probability of being within the repeatability value of each other.)

13.2.2 The (within laboratory) repeatability standard deviation is 1.92 °C. Therefore the repeatability value is 5.4 °C.

13.2.3 Between laboratory variability may be described using the reproducibility value (R) obtained by multiplying the reproducibility standard deviation by 2.8. The reproducibility value estimates the 95 % confidence limit. That is, two results obtained in different laboratories, or using different apparatus, or by different operators, or distant in time have a 95 % probability of being within the reproducibility value of each other.

13.2.4 The between laboratory reproducibility standard deviation is 3.85 °C. Therefore the reproducibility value is 10.8 °C.

13.3 *Bias:*

13.3.1 Bias is the difference between the determined value and that of an accepted reference value.

13.3.2 The mean value obtained in this study was 594.4 °C for a material whose value was taken to be 596 °C. The difference between the accepted value and the determined value is –1.6 °C. This difference is well within the reproducibility standard deviation indicating no evident bias.

14. Keywords

14.1 Curie point standards; magnetic transition standards; melting point standards; temperature calibration; thermogravimetry

[4] Supporting data have been filed at ASTM International Headquarters and may be obtained by requesting Research Report RR:E37-1017. Contact ASTM Customer Service at www.astm.org/contact.

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APPENDIX

(Nonmandatory Information)

X1. THERMAL LAG

X1.1 In TGA, the temperature sensor is not in direct contact with the test specimen leading to a difference in temperature between the sensor and the specimen that varies with heating and cooling rates. Temperature calibration must address this effect. One approach requires that the temperature calibration be performed at the same heating or cooling rate as that used for the test specimen.

X1.2 Temperature measurements performed at heating or cooling rates different from that of the calibration may not be accurate.

X1.3 The effect of heating or cooling rate on the temperature difference is known as thermal lag and is a characterisitic of the individual apparatus. Thermal lag may be assessed using Test Method E3142.

X1.4 The known thermal lag of an apparatus combined with Test Method E3142 may be used to adjust the temperature calibration obtained at one heating or cooling rate to that at another.

SUMMARY OF CHANGES

Committee E37 has identified the location of selected changes to this standard since the last issue (E1582 – 22) that may impact the use of this standard. (Approved Sep. 1, 2025.)

(1) Editorial updates throughout.
(2) Updates to 7.1.1.4, Note 2, and 9.1.
(3) Addition of Test Method E3142.
(4) Addition of Appendix X1.
(5) Removal of 9.2.4.

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